



DUBLIN CITY UNIVERSITY

SEMESTER 1 EXAMINATIONS 2015/2016

MODULE: CA208 - Logic

PROGRAMME(S):

CASE - BSc in Computer Applications (Sft.Eng.)
CPSSD - BSc in Computational Problem Solving and SW Dev.
ECSA - Study Abroad (Engineering and Computing)

YEAR OF STUDY: 2,X

EXAMINERS: Dr David Sinclair (Ph:5510)
Prof. David Bustard

TIME ALLOWED: 2 hours

INSTRUCTIONS: Answer ALL 6 questions from Section A. Answer 2 out of 3 questions from Section B. Question 1 carries 30 marks. Questions 2 to 9 carry 10 marks.

PLEASE DO NOT TURN OVER THIS PAGE UNTIL INSTRUCTED TO DO SO

The use of programmable or text storing calculators is expressly forbidden.
Please note that where a candidate answers more than the required number of questions, the examiner will mark all questions attempted and then select the highest scoring ones.

Requirements for this paper (Please mark (X) as appropriate)

☐	Log Tables
☐	Graph Paper
☐	Dictionaries
☐	Statistical Tables

☐	Thermodynamic Tables
☐	Actuarial Tables
☐	MCQ Only - Do not publish
☐	Attached Answer Sheet

Section A

QUESTION 1

[Total marks: 30]

All hounds howl at night. Anyone who has any cats will not have any mice. Light sleepers do not have anything that howls at night. John has either a cat or a hound.

1(a) [10 Marks]

Convert the statements above into a set of predicate formulae.

1(b) [10 Marks]

Convert the predicate formulae from Question 1(a) into *Prenex Conjunctive Normal Form (PCNF)*.

1(c) [10 Marks]

Using *General Resolution*, prove that if John is a light sleeper, then John does not have any mice.

[End Question 1]

QUESTION 2

[Total marks: 10]

[10 Marks]

Show that

$$(A \rightarrow C) \wedge (B \rightarrow C) \equiv (A \vee B) \rightarrow C$$

[End Question 2]

QUESTION 3

[Total marks: 10]

[10 Marks]

Using semantic tableaux, prove the following propositional formula is valid.

$$(p \rightarrow q) \rightarrow ((p \rightarrow \neg q) \rightarrow \neg p)$$

[End Question 3]

QUESTION 4**[Total marks: 10]**

[10 Marks]

Using the Gentzen System $\mathcal{G}$ prove

$$\vdash (A \rightarrow B) \rightarrow ((\neg A \rightarrow B) \rightarrow B)$$

[End Question 4]**QUESTION 5****[Total marks: 10]**

[10 Marks]

Using Gentzen System $\mathcal{G}$, prove the following predicate formula.

$$(\forall x.f(x) \rightarrow \exists x.g(x)) \rightarrow \exists x.(f(x) \rightarrow g(x))$$

[End Question 5]**QUESTION 6****[Total marks: 10]**

[10 Marks]

Convert the following formula to prenex conjunctive normal form (PCNF)

$$\exists x.\forall y.p(x, y) \rightarrow \forall y.\exists x.p(x, y)$$

[End Question 6]

Section B

QUESTION 7

[Total marks: 10]

[10 Marks]

Describe *cuts* in Prolog. Why are *cuts* used? Illustrate your answer with examples.

[End Question 7]

QUESTION 8

[Total marks: 10]

[10 Marks]

Write a Prolog relation *split(N, Positive, Negative)* that is true if all the positive numbers (including 0) in the list *N* are in the list *Positive*, and all the negative numbers in the list *N* are in the list *Negative*.

[End Question 8]

QUESTION 9

[Total marks: 10]

[10 Marks]

Let a set be represented by a list of elements. Write the following Prolog relations:

- *union(A,B,C)* is true if the list *C* contains all the elements in either lists *A* or *B*.
- *intersection(A,B,C)* is true if the list *C* contains all the elements in both lists *A* and *B*.

[End Question 9]

[APPENDICES]

Semantic Tableaux

α	α_1	α_2	β	β_1	β_2
$\neg\neg A_1$	A_1		$\neg(B_1 \wedge B_2)$	$\neg B_1$	$\neg B_2$
$A_1 \wedge A_2$	A_1	A_2	$B_1 \vee B_2$	B_1	B_2
$\neg(A_1 \vee A_2)$	$\neg A_1$	$\neg A_2$	$B_1 \rightarrow B_2$	$\neg B_1$	B_2
$\neg(A_1 \rightarrow A_2)$	A_1	$\neg A_2$	$B_1 \uparrow B_2$	$\neg B_1$	$\neg B_2$
$\neg(A_1 \uparrow A_2)$	A_1	A_2	$\neg(B_1 \downarrow B_2)$	B_1	B_2
$A_1 \downarrow A_2$	$\neg A_1$	$\neg A_2$	$\neg(B_1 \leftrightarrow B_2)$	$\neg(B_1 \rightarrow B_2)$	$\neg(B_2 \rightarrow B_1)$
$A_1 \leftrightarrow A_2$	$A_1 \rightarrow A_2$	$A_2 \rightarrow A_1$	$B_1 \oplus B_2$	$\neg(B_1 \rightarrow B_2)$	$\neg(B_2 \rightarrow B_1)$
$\neg(A_1 \oplus A_2)$	$A_1 \rightarrow A_2$	$A_2 \rightarrow A_1$			

γ	$\gamma(a)$	δ	$\delta(a)$
$\forall x.A(x)$	$A(a)$	$\exists x.A(x)$	$A(a)$
$\neg\exists x.A(x)$	$\neg A(a)$	$\neg\forall x.A(x)$	$\neg A(a)$

Gentzen Logic

$$\frac{\vdash U'_1 \cup \{\alpha_1, \alpha_2\}}{\vdash U'_1 \cup \{\alpha\}} \quad \frac{\vdash U'_1 \cup \{\beta_1\} \quad \vdash U'_2 \cup \{\beta_2\}}{\vdash U'_1 \cup U'_2 \cup \{\beta\}}$$

$$\frac{\vdash U'_1 \cup \{\gamma, \gamma(a)\}}{\vdash U'_1 \cup \{\gamma\}} \quad \frac{\vdash U'_1 \cup \{\delta(a)\}}{\vdash U'_1 \cup \{\delta\}}$$

$\frac{A}{B}$ is a *sequent* and means that if A is true, then B can be inferred.

α	α_1	α_2	β	β_1	β_2
$\neg\neg A$	A		$B_1 \wedge B_2$	B_1	B_2
$\neg(A_1 \wedge A_2)$	$\neg A_1$	$\neg A_2$	$\neg(B_1 \vee B_2)$	$\neg B_1$	$\neg B_2$
$A_1 \vee A_2$	A_1	A_2	$\neg(B_1 \rightarrow B_2)$	B_1	$\neg B_2$
$A_1 \rightarrow A_2$	$\neg A_1$	A_2	$\neg(B_1 \uparrow B_2)$	B_1	B_2
$A_1 \uparrow A_2$	$\neg A_1$	$\neg A_2$	$B_1 \downarrow B_2$	$\neg B_1$	$\neg B_2$
$\neg(A_1 \downarrow A_2)$	A_1	A_2	$B_1 \leftrightarrow B_2$	$B_1 \rightarrow B_2$	$B_2 \rightarrow B_1$
$\neg(A_1 \leftrightarrow A_2)$	$\neg(A_1 \rightarrow A_2)$	$\neg(A_2 \rightarrow A_1)$	$\neg(B_1 \oplus B_2)$	$B_1 \rightarrow B_2$	$B_2 \rightarrow B_1$
$A_1 \oplus A_2$	$\neg(A_1 \rightarrow A_2)$	$\neg(A_2 \rightarrow A_1)$			

γ	$\gamma(a)$	δ	$\delta(a)$
$\exists x.A(x)$	$A(a)$	$\forall x.A(x)$	$A(a)$
$\neg\forall x.A(x)$	$\neg A(a)$	$\neg\exists x.A(x)$	$\neg A(a)$

[END OF APPENDICES]

[END OF EXAM]